

10.1 The Potential Formulation

10.1.1 Scalar and Vector Potentials

In this chapter we seek the *general* solution to Maxwell's equations,

$$\left. \begin{array}{ll} \text{(i)} \quad \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho, & \text{(iii)} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \nabla \cdot \mathbf{B} = 0, & \text{(iv)} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{array} \right\} \quad (10.1)$$

Given the *sources* $\rho(\mathbf{r}, t)$ and $\mathbf{J}(\mathbf{r}, t)$, what are the *fields* $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B}(\mathbf{r}, t)$? In the static case, Coulomb's law and the Biot–Savart law provide the answer. What we're looking for, then, is the generalization of those laws to time-dependent configurations.

This is not an easy problem, and it pays to begin by representing the fields in terms of potentials. In electrostatics $\nabla \times \mathbf{E} = \mathbf{0}$ allowed us to write $\mathbf{E}$ as the gradient of a scalar potential: $\mathbf{E} = -\nabla V$. In *electrodynamics* this is no longer possible, because the curl of $\mathbf{E}$ is nonzero. But $\mathbf{B}$ remains divergenceless, so we can still write

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}}, \quad (10.2)$$

as in magnetostatics. Putting this into Faraday's law (iii) yields

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}),$$

or

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = \mathbf{0}.$$

Here is a quantity, unlike $\mathbf{E}$ alone, whose curl *does* vanish; it can therefore be written as the gradient of a scalar:

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla V.$$

In terms of V and $\mathbf{A}$, then,

$$\boxed{\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}}. \quad (10.3)$$

This reduces to the old form, of course, when $\mathbf{A}$ is constant.

The potential representation (Eqs. 10.2 and 10.3) automatically fulfills the two homogeneous Maxwell equations, (ii) and (iii). How about Gauss's law (i) and the Ampère/Maxwell law (iv)? Putting Eq. 10.3 into (i), we find that

$$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{1}{\epsilon_0} \rho; \quad (10.4)$$

this replaces Poisson's equation (to which it reduces in the static case). Putting Eqs. 10.2 and 10.3 into (iv) yields

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J} - \mu_0 \epsilon_0 \nabla \left(\frac{\partial V}{\partial t} \right) - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2},$$

or, using the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$, and rearranging the terms a bit,

$$(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2}) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) = -\mu_0 \mathbf{J}. \quad (10.5)$$

Equations 10.4 and 10.5 contain all the information in Maxwell's equations.

Example 10.1. Find the charge and current distributions that would give rise to the potentials

$$V = 0, \quad \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c} (ct - |x|)^2 \hat{\mathbf{z}} & \text{for } |x| < ct, \\ \mathbf{0} & \text{for } |x| > ct, \end{cases}$$

where k is a constant, and (of course) $c = 1/\sqrt{\epsilon_0 \mu_0}$.

Solution. First we'll determine the electric and magnetic fields, using Eqs. 10.2 and 10.3:

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 k}{2} (ct - |x|) \hat{\mathbf{z}},$$

$$\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\mu_0 k}{4c} \frac{\partial}{\partial x} (ct - |x|)^2 \hat{\mathbf{y}} = \pm \frac{\mu_0 k}{2c} (ct - |x|) \hat{\mathbf{y}}$$

(plus, for $x > 0$; minus, for $x < 0$). These are for $|x| < ct$; when $|x| > ct$, $\mathbf{E} = \mathbf{B} = \mathbf{0}$ (Fig. 10.1). Calculating every derivative in sight, I find

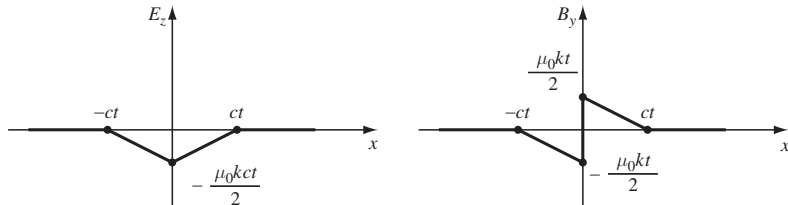


Fig. 10.1

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = \mp \frac{\mu_0 k}{2} \hat{\mathbf{y}}, \quad \nabla \times \mathbf{B} = -\frac{\mu_0 k}{2c} \hat{\mathbf{z}},$$

$$\frac{\partial \mathbf{E}}{\partial t} = -\frac{\mu_0 k c}{2} \hat{\mathbf{z}}, \quad \frac{\partial \mathbf{B}}{\partial t} = \pm \frac{\mu_0 k}{2} \hat{\mathbf{y}}.$$

As you can easily check, Maxwell's equations are all satisfied, with ρ and $\mathbf{J}$ both *zero*. Notice, however, that $\mathbf{B}$ has a discontinuity at $x = 0$, and this signals the presence of a surface current $\mathbf{K}$ in the yz plane; boundary condition (iv) in Eq. 7.73 gives

$$kt\hat{\mathbf{y}} = \mathbf{K} \times \hat{\mathbf{x}},$$

and hence

$$\mathbf{K} = kt\hat{\mathbf{z}}.$$

Evidently, we have here a uniform surface current flowing in the z direction over the plane $x = 0$, which starts up at $t = 0$, and increases in proportion to t . Notice that the “news” travels out (in both directions) at the speed of light: for points $|x| > ct$ the message (“current is now flowing”) has not yet arrived, so the fields are zero.

Problem 10.1. Show that the differential equations for V and $\mathbf{A}$ (Eqs. 10.4 and 10.5) can be written in the more symmetrical form

$$\left. \begin{aligned} \square^2 V + \frac{\partial L}{\partial t} &= -\frac{1}{\epsilon_0} \rho, \\ \square^2 \mathbf{A} - \nabla L &= -\mu_0 \mathbf{J}, \end{aligned} \right\} \quad (10.6)$$

where

$$\square^2 \equiv \nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t}.$$

Problem 10.2. For the configuration in Ex. 10.1, consider a rectangular box of length l , width w , and height h , situated a distance d above the yz plane (Fig. 10.2).

(a) Find the energy in the box at time $t_1 = d/c$, and at $t_2 = (d + h)/c$.

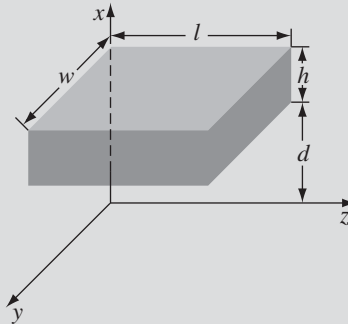


Fig. 10.2

- (b) Find the Poynting vector, and determine the energy per unit time flowing into the box during the interval $t_1 < t < t_2$.
- (c) Integrate the result in (b) from t_1 to t_2 , and confirm that the increase in energy (part (a)) equals the net influx.

10.1.2 Gauge Transformations

Equations 10.4 and 10.5 are *ugly*, and you might be inclined to abandon the potential formulation altogether. However, we *have* succeeded in reducing six problems – finding $\mathbf{E}$ and $\mathbf{B}$ (three components each) – down to four: V (one component) and $\mathbf{A}$ (three more). Moreover, Eqs. 10.2 and 10.3 do not uniquely define the potentials; we are free to impose extra conditions on V and $\mathbf{A}$, as long as nothing happens to $\mathbf{E}$ and $\mathbf{B}$. Let's work out precisely what this **gauge freedom** entails.

Suppose we have two sets of potentials, $(V, \mathbf{A})$ and $(V', \mathbf{A}')$, which correspond to the *same* electric and magnetic fields. By how much can they differ? Write

$$\mathbf{A}' = \mathbf{A} + \boldsymbol{\alpha} \quad \text{and} \quad V' = V + \beta.$$

Since the two $\mathbf{A}$ s give the same $\mathbf{B}$, their curls must be equal, and hence

$$\nabla \times \boldsymbol{\alpha} = \mathbf{0}.$$

We can therefore write $\boldsymbol{\alpha}$ as the gradient of a scalar:

$$\boldsymbol{\alpha} = \nabla \lambda.$$

The two potentials also give the same $\mathbf{E}$, so

$$\nabla \beta + \frac{\partial \boldsymbol{\alpha}}{\partial t} = \mathbf{0},$$

or

$$\nabla \left(\beta + \frac{\partial \lambda}{\partial t} \right) = \mathbf{0}.$$

The term in parentheses is therefore independent of position (it could, however, depend on time); call it $k(t)$:

$$\beta = -\frac{\partial \lambda}{\partial t} + k(t).$$

Actually, we might as well absorb $k(t)$ into λ , defining a new λ by adding $\int_0^t k(t') dt'$ to the old one. This will not affect the gradient of λ ; it just adds $k(t)$ to $\partial \lambda / \partial t$. It follows that

$$\left. \begin{aligned} \mathbf{A}' &= \mathbf{A} + \nabla \lambda, \\ V' &= V - \frac{\partial \lambda}{\partial t}. \end{aligned} \right\} \quad (10.7)$$

Conclusion: For any old scalar function $\lambda(\mathbf{r}, t)$, we can with impunity add $\nabla \lambda$ to $\mathbf{A}$, provided we simultaneously subtract $\partial \lambda / \partial t$ from V . This will

not affect the physical quantities $\mathbf{E}$ and $\mathbf{B}$. Such changes in V and $\mathbf{A}$ are called **gauge transformations**. They can be exploited to adjust the divergence of $\mathbf{A}$, with a view to simplifying the “ugly” equations 10.4 and 10.5. In magnetostatics, it was best to choose $\nabla \cdot \mathbf{A} = 0$ (Eq. 5.63); in electrodynamics, the situation is not so clear-cut, and the most convenient gauge depends to some extent on the problem at hand. There are many famous gauges in the literature; I’ll show you the two most popular ones.

Problem 10.3.

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\mathbf{r}, t) = 0, \quad \mathbf{A}(\mathbf{r}, t) = -\frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{\mathbf{r}}.$$

(b) Use the gauge function $\lambda = -(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Problem 10.4. Suppose $V = 0$ and $\mathbf{A} = A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$, where A_0 , ω , and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell’s equations in vacuum. What condition must you impose on ω and k ?

10.1.3 Coulomb Gauge and Lorenz Gauge

The Coulomb Gauge. As in magnetostatics, we pick

$$\nabla \cdot \mathbf{A} = 0. \quad (10.8)$$

With this, Eq. 10.4 becomes

$$\nabla^2 V = -\frac{1}{\epsilon_0} \rho. \quad (10.9)$$

This is Poisson’s equation, and we already know how to solve it: setting $V = 0$ at infinity,

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t)}{r} d\tau'. \quad (10.10)$$

There is a very peculiar thing about the scalar potential in the Coulomb gauge: it is determined by the distribution of charge *right now*. If I move an electron in my laboratory, the potential V on the moon immediately records this change. That sounds particularly odd in the light of special relativity, which allows no message to travel faster than c . The point is that V *by itself* is not a physically measurable quantity – all the man in the moon can measure is $\mathbf{E}$, and that involves $\mathbf{A}$ as well (Eq. 10.3). Somehow it is built into the vector potential (in the Coulomb gauge) that whereas V instantaneously

reflects all changes in ρ , the combination $-\nabla V - (\partial \mathbf{A}/\partial t)$ does *not*; $\mathbf{E}$ will change only after sufficient time has elapsed for the “news” to arrive.¹

The *advantage* of the Coulomb gauge is that the scalar potential is particularly simple to calculate; the *disadvantage* (apart from the acausal appearance of V) is that $\mathbf{A}$ is particularly *difficult* to calculate. The differential equation for $\mathbf{A}$ (Eq. 10.5) in the Coulomb gauge reads

$$\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J} + \mu_0 \epsilon_0 \nabla \left(\frac{\partial V}{\partial t} \right). \quad (10.11)$$

The Lorenz gauge. In the Lorenz² gauge, we pick

$$\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}. \quad (10.12)$$

This is designed to eliminate the middle term in Eq. 10.5 (in the language of Prob. 10.1, it sets $L = 0$). With this,

$$\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}. \quad (10.13)$$

Meanwhile, the differential equation for V (Eq. 10.4) becomes

$$\nabla^2 V - \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} = -\frac{1}{\epsilon_0} \rho. \quad (10.14)$$

The virtue of the Lorenz gauge is that it treats V and $\mathbf{A}$ on an equal footing: the same differential operator

$$\nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \equiv \square^2 \quad (10.15)$$

(called the **d'Alembertian**) occurs in both equations:

$$\begin{aligned} \text{(i)} \quad & \square^2 V = -\frac{1}{\epsilon_0} \rho, \\ \text{(ii)} \quad & \square^2 \mathbf{A} = -\mu_0 \mathbf{J}. \end{aligned} \quad (10.16)$$

This democratic treatment of V and $\mathbf{A}$ is especially nice in the context of special relativity, where the d'Alembertian is the natural generalization of the Laplacian, and Eqs. 10.16 can be regarded as four-dimensional versions of Poisson's equation. In this same spirit, the homogeneous wave equation $\square^2 f = 0$ might be regarded as the four-dimensional version of Laplace's equation. In the Lorenz gauge, V and $\mathbf{A}$ satisfy the **inhomogeneous wave equation**, with a “source” term (in place of zero) on the right. From now on, I shall use the Lorenz gauge exclusively, and the whole of electrodynamics

¹ See O. L. Brill and B. Goodman, *Am. J. Phys.* **35**, 832 (1967); J. D. Jackson, *Am. J. Phys.* **70**, 917 (2001).

² Until recently, it was spelled “Lorentz,” in honor of the Dutch physicist H. A. Lorentz, but it is now attributed to L. V. Lorenz, the Dane. See J. Van Bladel, *IEEE Antennas and Propagation Magazine* **33**(2), 69 (1991); J. D. Jackson and L. B. Okun, *Rev. Mod. Phys.* **73**, 663 (2001).

reduces to the problem of *solving the inhomogeneous wave equation for a specified source*.

Problem 10.5. Which of the potentials in Ex. 10.1, Prob. 10.3, and Prob. 10.4 are in the Coulomb gauge? Which are in the Lorenz gauge? (Notice that these gauges are not mutually exclusive.)

Problem 10.6. In Chapter 5, I showed that it is always possible to pick a vector potential whose divergence is zero (the Coulomb gauge). Show that it is always possible to choose $\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 (\partial V / \partial t)$, as required for the Lorenz gauge, assuming you know how to solve the inhomogeneous wave equation (Eq. 10.16). Is it always possible to pick $V = 0$? How about $\mathbf{A} = \mathbf{0}$?

Problem 10.7. A time-dependent point charge $q(t)$ at the origin, $\rho(\mathbf{r}, t) = q(t)\delta^3(\mathbf{r})$, is fed by a current $\mathbf{J}(\mathbf{r}, t) = -(1/4\pi)(\dot{q}/r^2)\hat{\mathbf{r}}$, where $\dot{q} \equiv dq/dt$.

- Check that charge is conserved, by confirming that the continuity equation is obeyed.
- Find the scalar and vector potentials in the Coulomb gauge. If you get stuck, try working on (c) first.
- Find the fields, and check that they satisfy all of Maxwell's equations.³

10.1.4 Lorentz Force Law in Potential Form

It⁴ is illuminating to express the Lorentz force law in terms of potentials:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q\left[-\nabla V - \frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \times (\nabla \times \mathbf{A})\right], \quad (10.17)$$

where $\mathbf{p} = m\mathbf{v}$ is the momentum of the particle. Now, product rule 4 says

$$\nabla(\mathbf{v} \cdot \mathbf{A}) = \mathbf{v} \times (\nabla \times \mathbf{A}) + (\mathbf{v} \cdot \nabla)\mathbf{A}$$

($\mathbf{v}$, the velocity of the particle, is a function of time, but not of position). Thus

$$\frac{d\mathbf{p}}{dt} = -q\left[\frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A} + \nabla(V - \mathbf{v} \cdot \mathbf{A})\right]. \quad (10.18)$$

The combination

$$\left[\frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A}\right]$$

is called the **convective derivative** of $\mathbf{A}$, and written $d\mathbf{A}/dt$ (*total derivative*). It represents the time rate of change of $\mathbf{A}$ *at the (moving) location of the particle*. For suppose that at time t the particle is at point $\mathbf{r}$, where the potential is $\mathbf{A}(\mathbf{r}, t)$; a moment dt later it is at $\mathbf{r} + \mathbf{v} dt$, where the potential is $\mathbf{A}(\mathbf{r} + \mathbf{v} dt, t + dt)$. The *change* in $\mathbf{A}$, then, is

³ P. R. Berman, *Am. J. Phys.* **76**, 48 (2008).

⁴ This section can be skipped without loss of continuity.

$$\begin{aligned}
 d\mathbf{A} &= \mathbf{A}(\mathbf{r} + \mathbf{v} dt, t + dt) - \mathbf{A}(\mathbf{r}, t) \\
 &= \left(\frac{\partial \mathbf{A}}{\partial x}\right)(v_x dt) + \left(\frac{\partial \mathbf{A}}{\partial y}\right)(v_y dt) + \left(\frac{\partial \mathbf{A}}{\partial z}\right)(v_z dt) + \left(\frac{\partial \mathbf{A}}{\partial t}\right) dt,
 \end{aligned}$$

so

$$\frac{d\mathbf{A}}{dt} = \frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A}. \quad (10.19)$$

As the particle moves, the potential it “feels” changes for two distinct reasons: first, because the potential varies with *time*, and second, because it is now in a new location, where $\mathbf{A}$ is different because of its variation in *space*. Hence the two terms in Eq. 10.19.

With the aid of the convective derivative, the Lorentz force law reads:

$$\frac{d}{dt}(\mathbf{p} + q\mathbf{A}) = -\nabla[q(V - \mathbf{v} \cdot \mathbf{A})]. \quad (10.20)$$

This is reminiscent of the standard formula from mechanics, for the motion of a particle whose potential energy U is a specified function of position:

$$\frac{d\mathbf{p}}{dt} = -\nabla U.$$

Playing the role of $\mathbf{p}$ is the so-called **canonical momentum**,

$$\mathbf{p}_{\text{can}} = \mathbf{p} + q\mathbf{A}, \quad (10.21)$$

while the part of U is taken by the velocity-dependent quantity

$$U_{\text{vel}} = q(V - \mathbf{v} \cdot \mathbf{A}). \quad (10.22)$$

A similar argument (Prob. 10.9) gives the rate of change of the particle’s *energy*:

$$\frac{d}{dt}(T + qV) = \frac{\partial}{\partial t}[q(V - \mathbf{v} \cdot \mathbf{A})], \quad (10.23)$$

where $T = \frac{1}{2}mv^2$ is its kinetic energy and qV is its potential energy (the derivative on the right acts only on V and $\mathbf{A}$, not on $\mathbf{v}$). Curiously, the same quantity⁵ U_{vel} appears on the right-hand side of both equations. The parallel between Eq. 10.20 and Eq. 10.23 invites us to interpret $\mathbf{A}$ as a kind of “potential momentum” per unit charge, just as V is potential *energy* per unit charge.⁶

Problem 10.8. The vector potential for a uniform magnetostatic field is $\mathbf{A} = -\frac{1}{2}(\mathbf{r} \times \mathbf{B})$ (Prob. 5.27). Show that $d\mathbf{A}/dt = -\frac{1}{2}(\mathbf{v} \times \mathbf{B})$, in this case, and confirm that Eq. 10.20 yields the correct equation of motion.

Problem 10.9. Derive Eq. 10.23. [*Hint*: Start by dotting $\mathbf{v}$ into Eq. 10.17.]

⁵ I don’t know what to call U_{vel} – it’s not potential energy, exactly (that would be qV).

⁶ There are other arguments for this interpretation, which Maxwell himself favored, and many modern authors advocate. For a fascinating discussion, see M. D. Semon and J. R. Taylor, *Am. J. Phys.* **64**, 1361 (1996). Incidentally, it is the *canonical* angular momentum (derived from $\mathbf{p}_{\text{can}}$), not the mechanical portion alone, that is quantized – see R. H. Young, *Am. J. Phys.* **66**, 1043 (1998).

10.2 Continuous Distributions

10.2.1 Retarded Potentials

In the static case, Eq. 10.16 reduces to (four copies of) Poisson's equation,

$$\nabla^2 V = -\frac{1}{\epsilon_0} \rho, \quad \nabla^2 \mathbf{A} = -\mu_0 \mathbf{J},$$

with the familiar solutions

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau', \quad \mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau', \quad (10.24)$$

where r , as always, is the distance from the source point $\mathbf{r}'$ to the field point $\mathbf{r}$ (Fig. 10.3). Now, electromagnetic “news” travels at the speed of light. In the *nonstatic* case, therefore, it's not the status of the source *right now* that matters, but rather its condition at some earlier time t_r (called the **retarded time**) when the “message” left. Since this message must travel a distance r , the delay is r/c :

$$t_r \equiv t - \frac{r}{c}. \quad (10.25)$$

The natural generalization of Eq. 10.24 for *nonstatic* sources is therefore

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{r} d\tau', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{r} d\tau'. \quad (10.26)$$

Here $\rho(\mathbf{r}', t_r)$ is the charge density that prevailed at point $\mathbf{r}'$ at the retarded time t_r . Because the integrands are evaluated at the retarded time, these are called **retarded potentials**. (I speak of “the” retarded time, but of course more distant parts of the charge distribution have earlier retarded times than nearby ones. It's just like the night sky: the light we see now left each star at the retarded time corresponding to that star's distance from the Earth.) Note that the retarded potentials reduce properly to Eq. 10.24 in the static case, for which ρ and $\mathbf{J}$ are independent of time.

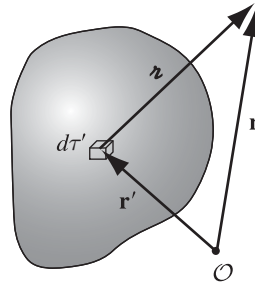


Fig. 10.3

Well, that all sounds *reasonable* – and surprisingly simple. But are we sure it's *right*? I didn't actually *derive* the formulas for V and $\mathbf{A}$ (Eq. 10.26); all I did was invoke a heuristic argument ("electromagnetic news travels at the speed of light") to make them seem *plausible*. To *prove* them, I must show that they satisfy the inhomogeneous wave equation (Eq. 10.16) and meet the Lorenz condition (Eq. 10.12). In case you think I'm being fussy, let me warn you that if you apply the same logic to the *fields* you'll get entirely the *wrong* answer (see Prob. 10.36 for the correct argument):

$$\mathbf{E}(\mathbf{r}, t) \neq \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{r^2} \hat{\mathbf{r}} d\tau', \quad \mathbf{B}(\mathbf{r}, t) \neq \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r) \times \hat{\mathbf{r}}}{r^2} d\tau'.$$

Let's stop and check, then, that the retarded scalar potential satisfies Eq. 10.16; essentially the same argument would serve for the vector potential.⁷ I shall leave it for you (Prob. 10.10) to show that the retarded potentials obey the Lorenz condition.

Proof. In calculating the Laplacian of $V(\mathbf{r}, t)$, the crucial point to notice is that the integrand (in Eq. 10.26) depends on $\mathbf{r}$ in *two* places: *explicitly*, in the denominator ($r = |\mathbf{r} - \mathbf{r}'|$), and *implicitly*, through $t_r = t - r/c$, in the numerator. Thus

$$\nabla V = \frac{1}{4\pi\epsilon_0} \int \left[(\nabla \rho) \frac{1}{r} + \rho \nabla \left(\frac{1}{r} \right) \right] d\tau', \quad (10.27)$$

and

$$\nabla \rho = \dot{\rho} \nabla t_r = -\frac{1}{c} \dot{\rho} \nabla r. \quad (10.28)$$

(The dot denotes differentiation with respect to time).⁸ Now $\nabla r = \hat{\mathbf{r}}$ and $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$ (Prob. 1.13), so

$$\nabla V = \frac{1}{4\pi\epsilon_0} \int \left[-\frac{\dot{\rho}}{c} \frac{\hat{\mathbf{r}}}{r} - \rho \frac{\hat{\mathbf{r}}}{r^2} \right] d\tau'. \quad (10.29)$$

Taking the divergence,

$$\begin{aligned} \nabla^2 V = \frac{1}{4\pi\epsilon_0} \int \left\{ -\frac{1}{c} \left[\frac{\hat{\mathbf{r}}}{r} \cdot (\nabla \dot{\rho}) + \dot{\rho} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r} \right) \right] \right. \\ \left. - \left[\frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla \rho) + \rho \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \right] \right\} d\tau'. \end{aligned}$$

But

$$\nabla \dot{\rho} = -\frac{1}{c} \ddot{\rho} \nabla r = -\frac{1}{c} \ddot{\rho} \hat{\mathbf{r}},$$

as in Eq. 10.28, and

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r} \right) = \frac{1}{r^2}$$

⁷ I'll give you the straightforward but cumbersome proof; for a clever indirect argument see M. A. Heald and J. B. Marion, *Classical Electromagnetic Radiation*, 3rd ed., Saunders, Orlando, FL (1995), Sect. 8.1.

⁸ Note that $\partial/\partial t_r = \partial/\partial t$, since $t_r = t - r/c$ and r is independent of t .

(Prob. 1.63), whereas

$$\nabla \cdot \left(\frac{\hat{\mathbf{z}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r})$$

(Eq. 1.100). So

$$\nabla^2 V = \frac{1}{4\pi\epsilon_0} \int \left[\frac{1}{c^2} \frac{\ddot{\rho}}{r} - 4\pi\rho\delta^3(\mathbf{r}) \right] d\tau' = \frac{1}{c^2} \frac{\partial^2 V}{\partial t^2} - \frac{1}{\epsilon_0} \rho(\mathbf{r}, t),$$

confirming that the retarded potential (Eq. 10.26) satisfies the inhomogeneous wave equation (Eq. 10.16). $\square$

Incidentally, this proof applies equally well to the **advanced potentials**,

$$V_a(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_a)}{r} d\tau', \quad \mathbf{A}_a(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_a)}{r} d\tau', \quad (10.30)$$

in which the charge and the current densities are evaluated at the **advanced time**

$$t_a \equiv t + \frac{r}{c}. \quad (10.31)$$

A few signs are changed, but the final result is unaffected. Although the advanced potentials are entirely consistent with Maxwell's equations, they violate the most sacred tenet in all of physics: the principle of **causality**. They suggest that the potentials *now* depend on what the charge and the current distribution *will be* at some time in the future – the effect, in other words, precedes the cause. Although the advanced potentials are of some theoretical interest, they have no direct physical significance.⁹

Example 10.2. An infinite straight wire carries the current

$$I(t) = \begin{cases} 0 & \text{for } t \leq 0, \\ I_0 & \text{for } t > 0. \end{cases}$$

That is, a constant current I_0 is turned on abruptly at $t=0$. Find the resulting electric and magnetic fields.

Solution. The wire is presumably electrically neutral, so the scalar potential is zero. Let the wire lie along the z -axis (Fig. 10.4); the retarded vector potential at point P is

$$\mathbf{A}(s, t) = \frac{\mu_0}{4\pi} \hat{\mathbf{z}} \int_{-\infty}^{\infty} \frac{I(t_r)}{r} dz.$$

For $t < s/c$, the “news” has not yet reached P , and the potential is zero. For $t > s/c$, only the segment

⁹ Because the d'Alembertian involves t^2 (as opposed to t), the theory itself is **time-reversal invariant**, and does not distinguish “past” from “future.” Time asymmetry is introduced when we select the retarded potentials in preference to the advanced ones, reflecting the (not unreasonable!) belief that electromagnetic influences propagate forward, not backward, in time.

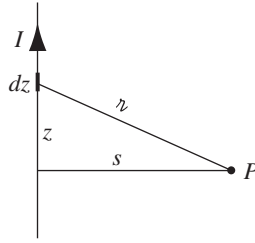


Fig. 10.4

$$|z| \leq \sqrt{(ct)^2 - s^2} \quad (10.32)$$

contributes (outside this range t_r is negative, so $I(t_r) = 0$); thus

$$\begin{aligned} \mathbf{A}(s, t) &= \left(\frac{\mu_0 I_0}{4\pi} \hat{\mathbf{z}} \right) 2 \int_0^{\sqrt{(ct)^2 - s^2}} \frac{dz}{\sqrt{s^2 + z^2}} \\ &= \frac{\mu_0 I_0}{2\pi} \hat{\mathbf{z}} \ln(\sqrt{s^2 + z^2} + z) \Big|_0^{\sqrt{(ct)^2 - s^2}} = \frac{\mu_0 I_0}{2\pi} \ln \left(\frac{ct + \sqrt{(ct)^2 - s^2}}{s} \right) \hat{\mathbf{z}}. \end{aligned}$$

The electric field is

$$\mathbf{E}(s, t) = -\frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 I_0 c}{2\pi \sqrt{(ct)^2 - s^2}} \hat{\mathbf{z}},$$

and the magnetic field is

$$\mathbf{B}(s, t) = \nabla \times \mathbf{A} = -\frac{\partial A_z}{\partial s} \hat{\phi} = \frac{\mu_0 I_0}{2\pi s} \frac{ct}{\sqrt{(ct)^2 - s^2}} \hat{\phi}.$$

Notice that as $t \rightarrow \infty$ we recover the static case: $\mathbf{E} = \mathbf{0}$, $\mathbf{B} = (\mu_0 I_0 / 2\pi s) \hat{\phi}$.

! **Problem 10.10.** Confirm that the retarded potentials satisfy the Lorenz gauge condition. [Hint: First show that

$$\nabla \cdot \left(\frac{\mathbf{J}}{r} \right) = \frac{1}{r} (\nabla \cdot \mathbf{J}) + \frac{1}{r} (\nabla' \cdot \mathbf{J}) - \nabla' \cdot \left(\frac{\mathbf{J}}{r} \right),$$

where ∇ denotes derivatives with respect to $\mathbf{r}$, and ∇' denotes derivatives with respect to $\mathbf{r}'$. Next, noting that $\mathbf{J}(\mathbf{r}', t - r/c)$ depends on $\mathbf{r}'$ both explicitly and through r , whereas it depends on $\mathbf{r}$ only through r , confirm that

$$\nabla \cdot \mathbf{J} = -\frac{1}{c} \dot{\mathbf{J}} \cdot (\nabla r), \quad \nabla' \cdot \mathbf{J} = -\dot{\rho} - \frac{1}{c} \dot{\mathbf{J}} \cdot (\nabla' r).$$

Use this to calculate the divergence of $\mathbf{A}$ (Eq. 10.26).]

! **Problem 10.11.**

(a) Suppose the wire in Ex. 10.2 carries a linearly increasing current

$$I(t) = kt,$$

for $t > 0$. Find the electric and magnetic fields generated.

(b) Do the same for the case of a sudden burst of current:

$$I(t) = q_0 \delta(t).$$

Problem 10.12. A piece of wire bent into a loop, as shown in Fig. 10.5, carries a current that increases linearly with time:

$$I(t) = kt \quad (-\infty < t < \infty).$$

Calculate the retarded vector potential $\mathbf{A}$ at the center. Find the electric field at the center. Why does this (neutral) wire produce an *electric* field? (Why can't you determine the *magnetic* field from this expression for $\mathbf{A}$?)

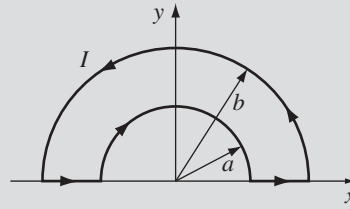


Fig. 10.5

10.2.2 Jefimenko's Equations

Given the retarded potentials

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{r} d\tau', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{r} d\tau', \quad (10.33)$$

it is, in principle, a straightforward matter to determine the fields:

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (10.34)$$

But the details are not entirely trivial because, as I mentioned earlier, the integrands depend on $\mathbf{r}$ both explicitly, through $r = |\mathbf{r} - \mathbf{r}'|$ in the denominator, and implicitly, through the retarded time $t_r = t - r/c$ in the argument of the numerator.

I already calculated the gradient of V (Eq. 10.29); the time derivative of $\mathbf{A}$ is easy:

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{\mu_0}{4\pi} \int \frac{\dot{\mathbf{J}}}{r} d\tau'. \quad (10.35)$$

Putting them together (and using $c^2 = 1/\mu_0\epsilon_0$),

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \left[\frac{\rho(\mathbf{r}', t_r)}{r^2} \hat{\mathbf{r}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{c r} \hat{\mathbf{r}} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2 r} \right] d\tau'. \quad (10.36)$$

This is the time-dependent generalization of Coulomb's law, to which it reduces in the static case (where the second and third terms drop out and the first term loses its dependence on t_r).

As for $\mathbf{B}$, the curl of $\mathbf{A}$ contains two terms:

$$\nabla \times \mathbf{A} = \frac{\mu_0}{4\pi} \int \left[\frac{1}{\mathcal{L}} (\nabla \times \mathbf{J}) - \mathbf{J} \times \nabla \left(\frac{1}{\mathcal{L}} \right) \right] d\tau'.$$

Now

$$(\nabla \times \mathbf{J})_x = \frac{\partial J_z}{\partial y} - \frac{\partial J_y}{\partial z},$$

and

$$\frac{\partial J_z}{\partial y} = j_z \frac{\partial t_r}{\partial y} = -\frac{1}{c} j_z \frac{\partial \mathcal{L}}{\partial y},$$

so

$$(\nabla \times \mathbf{J})_x = -\frac{1}{c} \left(j_z \frac{\partial \mathcal{L}}{\partial y} - j_y \frac{\partial \mathcal{L}}{\partial z} \right) = \frac{1}{c} [\mathbf{j} \times (\nabla \mathcal{L})]_x.$$

But $\nabla \mathcal{L} = \hat{\mathbf{r}}$ (Prob. 1.13), so

$$\nabla \times \mathbf{J} = \frac{1}{c} \mathbf{j} \times \hat{\mathbf{r}}. \quad (10.37)$$

Meanwhile, $\nabla(1/\mathcal{L}) = -\hat{\mathbf{r}}/\mathcal{L}^2$ (again, Prob. 1.13), and hence

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{\mathcal{L}^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c\mathcal{L}} \right] \times \hat{\mathbf{r}} d\tau'.} \quad (10.38)$$

This is the time-dependent generalization of the Biot–Savart law, to which it reduces in the static case.

Equations 10.36 and 10.38 are the (causal) solutions to Maxwell's equations. For some reason, they do not seem to have been published until quite recently – the earliest explicit statement of which I am aware was by Oleg Jefimenko, in 1966.¹⁰ In practice **Jefimenko's equations** are of limited utility, since it is typically easier to calculate the retarded potentials and differentiate them, rather than going directly to the fields. Nevertheless, they provide a satisfying sense of closure to the theory. They also help to clarify an observation I made in the previous section: to get to the retarded *potentials*, all you do is replace t by t_r in the electrostatic and magnetostatic formulas, but in the case of the *fields* not only is time replaced by retarded time, but completely new terms (involving derivatives of ρ and $\mathbf{J}$) appear. And Jefimenko's equations provide surprisingly strong support for the quasistatic approximation (see Prob. 10.14).

¹⁰ O. D. Jefimenko, *Electricity and Magnetism*, Appleton-Century-Crofts, New York (1966), Sect. 15.7. Related expressions appear in G. A. Schott, *Electromagnetic Radiation*, Cambridge University Press, Cambridge, UK (1912), Chap. 2; W. K. H. Panofsky and M. Phillips, *Classical Electricity and Magnetism*, Addison-Wesley, Reading, MA (1962), Sect. 14.3, and elsewhere. See K. T. McDonald, *Am. J. Phys.* **65**, 1074 (1997) for illuminating commentary and references.

Problem 10.13. Suppose $\mathbf{J}(\mathbf{r})$ is constant in time, so (Prob. 7.63) $\rho(\mathbf{r}, t) = \rho(\mathbf{r}, 0) + \dot{\rho}(\mathbf{r}, 0)t$. Show that

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t)}{r^2} \hat{\mathbf{z}} d\tau';$$

that is, Coulomb's law holds, with the charge density evaluated at the *nonretarded* time.

Problem 10.14. Suppose the current density changes slowly enough that we can (to good approximation) ignore all higher derivatives in the Taylor expansion

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t) + \dots$$

(for clarity, I suppress the $\mathbf{r}$ -dependence, which is not at issue). Show that a fortuitous cancellation in Eq. 10.38 yields

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t) \times \hat{\mathbf{z}}}{r^2} d\tau'.$$

That is, the Biot–Savart law holds, with $\mathbf{J}$ evaluated at the *nonretarded* time. This means that the quasistatic approximation is actually much *better* than we had any right to expect: the *two* errors involved (neglecting retardation and dropping the second term in Eq. 10.38) *cancel* one another, to first order.

! **Problem 10.15.** Obtain the fields for Ex. 10.2 directly from Jefimenko's equations.¹¹

10.3 Point Charges

10.3.1 Liénard–Wiechert Potentials

My next project is to calculate the (retarded) potentials, $V(\mathbf{r}, t)$ and $\mathbf{A}(\mathbf{r}, t)$, of a point charge q that is moving on a specified trajectory

$$\mathbf{w}(t) \equiv \text{position of } q \text{ at time } t. \quad (10.39)$$

A naïve reading of the formula (Eq. 10.26)

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{r} d\tau' \quad (10.40)$$

might suggest to you that the potential is simply

$$\frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

(the same as in the static case, with the understanding that r is the distance to the *retarded* position of the charge). But this is wrong, for a very subtle

¹¹ D. V. Redžić and V. Hnizdo, *Eur. J. Phys.* **34**, 495 (2013).

reason: it is true that for a point source the denominator z comes outside the integral,¹² but what remains,

$$\int \rho(\mathbf{r}', t_r) d\tau', \quad (10.41)$$

is *not* equal to the charge of the particle (and depends, through t_r , on the location of the point $\mathbf{r}$). To calculate the total charge of a configuration, you must integrate ρ over the entire distribution at *one instant of time*, but here the retardation, $t_r = t - z/c$, obliges us to evaluate ρ at *different times* for different parts of the configuration. If the source is moving, the result will *not* be the total charge. You might think that this problem would disappear for *point* charges, but it doesn't. In Maxwell's electrodynamics, formulated as it is in terms of charge and current *densities*, a point charge must be regarded as the limit of an extended charge, when the size goes to zero. And for an extended particle, no matter how small, the retardation in Eq. 10.41 throws in a factor $(1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c)^{-1}$, where $\mathbf{v}$ is the velocity of the charge at the retarded time:

$$\int \rho(\mathbf{r}', t_r) d\tau' = \frac{q}{1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c}. \quad (10.42)$$

Proof. This is a purely *geometrical* effect, and it may help to tell the story in a less abstract context. You will not have noticed it, for obvious reasons, but the fact is that a train coming toward you looks a little longer than it really is, because the light you receive from the caboose left earlier than the light you receive (at the same instant) from the engine, and at that earlier time the train was farther away (Fig. 10.6). In the interval it takes light from the caboose to travel that longer distance L' , the train itself moves a distance $L' - L$:

$$\frac{L'}{c} = \frac{L' - L}{v}, \quad \text{or } L' = \frac{L}{1 - v/c}.$$

So approaching trains appear *longer*, by a factor $(1 - v/c)^{-1}$. By contrast, a train going *away* from you looks *shorter*,¹³ by a factor $(1 + v/c)^{-1}$. In general, if the train's velocity makes an angle θ with

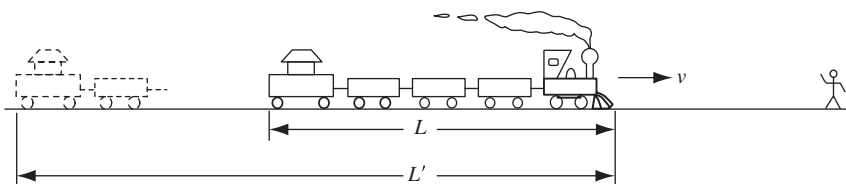


Fig. 10.6

¹² There is, however, an implicit change in its functional dependence: *before* the integration, $z = |\mathbf{r} - \mathbf{r}'|$ is a function of $\mathbf{r}$ and $\mathbf{r}'$; *after* the integration, which fixes $\mathbf{r}' = \mathbf{w}(t_r)$, $z = |\mathbf{r} - \mathbf{w}(t_r)|$ is (like t_r) a function of $\mathbf{r}$ and t .

¹³ Please note that this has nothing whatever to do with special relativity or Lorentz contraction – L is the length of the *moving* train, and its *rest* length is not at issue. The argument is somewhat reminiscent of the Doppler effect.

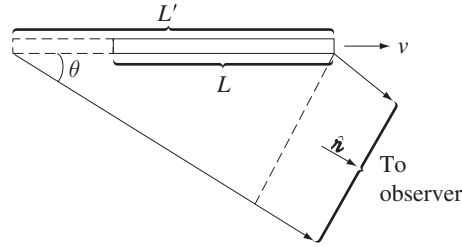


Fig. 10.7

your line of sight,¹⁴ the extra distance light from the caboose must cover is $L' \cos \theta$ (Fig. 10.7). In the time $L' \cos \theta/c$, then, the train moves a distance $(L' - L)$:

$$\frac{L' \cos \theta}{c} = \frac{L' - L}{v}, \quad \text{or } L' = \frac{L}{1 - v \cos \theta/c}.$$

Notice that this effect does *not* distort the dimensions perpendicular to the motion (the height and width of the train). Never mind that the light from the far side is delayed in reaching you (relative to light from the near side) – since there's no *motion* in that direction, they'll still look the same distance apart. The apparent *volume* τ' of the train, then, is related to the *actual* volume τ by

$$\tau' = \frac{\tau}{1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c}, \quad (10.43)$$

where $\hat{\mathbf{z}}$ is a unit vector from the train to the observer.

In case the connection between moving trains and retarded potentials eludes you, the point is this: whenever you do an integral of the type in Eq. 10.41, in which the integrand is evaluated at the *retarded* time, the effective volume is not the same as the *actual* volume; rather, it is the latter modified by the factor in Eq. 10.43, just as the apparent volume of the train was. Because this correction factor makes no reference to the *size* of the particle, it is every bit as significant for a point charge as for an extended charge. $\square$

Meanwhile, for a point charge the retarded time is determined implicitly by the equation

$$|\mathbf{r} - \mathbf{w}(t_r)| = c(t - t_r). \quad (10.44)$$

The left-hand side is the distance the “news” must travel, and $(t - t_r)$ is the time it takes to make the trip (Fig. 10.8); $\mathbf{z}$ is the vector from the retarded position to the field point $\mathbf{r}$:

$$\mathbf{z} = \mathbf{r} - \mathbf{w}(t_r). \quad (10.45)$$

¹⁴ I assume the train is far enough away or (more to the point) *short* enough so that rays from the caboose and engine can be considered parallel.

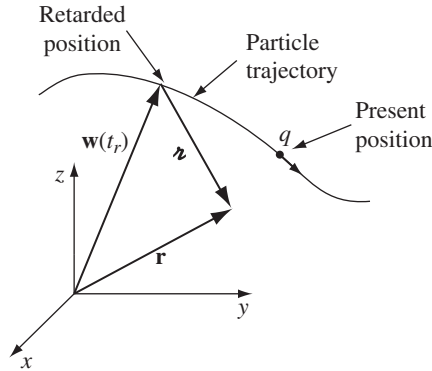


Fig. 10.8

It is important to note that at most *one* point on the trajectory is “in communication” with $\mathbf{r}$ at any particular time t . For suppose there were *two* such points, with retarded times t_1 and t_2 :

$$r_1 = c(t - t_1) \quad \text{and} \quad r_2 = c(t - t_2).$$

Then $r_1 - r_2 = c(t_2 - t_1)$, so the average speed of the particle in the direction of the point $\mathbf{r}$ would have to be c – and that’s not counting whatever velocity the charge might have in *other* directions. Since no charged particle can travel at the speed of light, it follows that *only one retarded point contributes to the potentials, at any given moment.*¹⁵

It follows, then, that

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{(rc - \mathbf{z} \cdot \mathbf{v})}, \quad (10.46)$$

where $\mathbf{v}$ is the velocity of the charge at the retarded time, and $\mathbf{z}$ is the vector from the retarded position to the field point $\mathbf{r}$. Moreover, since the current density is $\rho\mathbf{v}$ (Eq. 5.26), the vector potential is

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\rho(\mathbf{r}', t_r)\mathbf{v}(t_r)}{r} d\tau' = \frac{\mu_0}{4\pi} \frac{\mathbf{v}}{r} \int \rho(\mathbf{r}', t_r) d\tau',$$

or

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \frac{qc\mathbf{v}}{(rc - \mathbf{z} \cdot \mathbf{v})} = \frac{\mathbf{v}}{c^2} V(\mathbf{r}, t). \quad (10.47)$$

Equations 10.46 and 10.47 are the famous **Liénard–Wiechert potentials** for a moving point charge.¹⁶

¹⁵ For the same reason, an observer at $\mathbf{r}$ *sees* the particle in only one place at a time. By contrast, it is possible to *hear* an object in two places at once. Consider a bear who growls at you and then runs toward you at the speed of sound and growls again; you hear both growls at the same time, coming from two different locations, but there’s only one bear.

¹⁶ There are many ways to obtain the Liénard–Wiechert potentials. I have tried to emphasize the *geometrical* origin of the factor $(1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c)^{-1}$; for illuminating commentary, see W. K. H. Panofsky and M. Phillips, *Classical Electricity and Magnetism*, 2nd ed., Addison-Wesley,

Example 10.3. Find the potentials of a point charge moving with constant velocity.

Solution. For convenience, let's say the particle passes through the origin at time $t = 0$, so that

$$\mathbf{w}(t) = \mathbf{v}t.$$

We first compute the retarded time, using Eq. 10.44:

$$|\mathbf{r} - \mathbf{v}t_r| = c(t - t_r),$$

or, squaring:

$$r^2 - 2\mathbf{r} \cdot \mathbf{v}t_r + v^2 t_r^2 = c^2(t^2 - 2tt_r + t_r^2).$$

Solving for t_r by the quadratic formula, we find that

$$t_r = \frac{(c^2 t - \mathbf{r} \cdot \mathbf{v}) \pm \sqrt{(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)}}{c^2 - v^2}. \quad (10.48)$$

To fix the sign, consider the limit $v = 0$:

$$t_r = t \pm \frac{r}{c}.$$

In this case the charge is at rest at the origin, and the retarded time should be $(t - r/c)$; evidently we want the *minus* sign.

Now, from Eqs. 10.44 and 10.45,

$$\mathcal{r} = c(t - t_r) \quad \text{and} \quad \hat{\mathbf{r}} = \frac{\mathbf{r} - \mathbf{v}t_r}{c(t - t_r)},$$

so

$$\begin{aligned} \mathcal{r}(1 - \hat{\mathbf{r}} \cdot \mathbf{v}/c) &= c(t - t_r) \left[1 - \frac{\mathbf{v}}{c} \cdot \frac{(\mathbf{r} - \mathbf{v}t_r)}{c(t - t_r)} \right] = c(t - t_r) - \frac{\mathbf{v} \cdot \mathbf{r}}{c} + \frac{v^2}{c} t_r \\ &= \frac{1}{c} [(c^2 t - \mathbf{r} \cdot \mathbf{v}) - (c^2 - v^2)t_r] \\ &= \frac{1}{c} \sqrt{(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)} \end{aligned}$$

(I used Eq. 10.48, with the minus sign, in the last step). Therefore,

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{\sqrt{(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)}}, \quad (10.49)$$

and (Eq. 10.47)

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \frac{qc\mathbf{v}}{\sqrt{(c^2 t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2 t^2)}}. \quad (10.50)$$

Reading, MA (1962), pp. 342–343. A more rigorous derivation is provided by J. R. Reitz, F. J. Milford, and R. W. Christy, *Foundations of Electromagnetic Theory*, 3rd ed., Addison-Wesley, Reading, MA (1979), Sect. 21.1, or M. A. Heald and J. B. Marion, *Classical Electromagnetic Radiation*, 3rd ed., Saunders, Orlando, FL (1995), Sect. 8.3.

Problem 10.16. A particle of charge q moves in a circle of radius a at constant angular velocity ω . (Assume that the circle lies in the xy plane, centered at the origin, and at time $t = 0$ the charge is at $(a, 0)$, on the positive x -axis.) Find the Liénard–Wiechert potentials for points on the z -axis.

- **Problem 10.17.** Show that the scalar potential of a point charge moving with constant velocity (Eq. 10.49) can be written more simply as

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{q}{R\sqrt{1 - v^2 \sin^2 \theta/c^2}}, \quad (10.51)$$

where $\mathbf{R} \equiv \mathbf{r} - \mathbf{v}t$ is the vector from the *present(!)* position of the particle to the field point $\mathbf{r}$, and θ is the angle between $\mathbf{R}$ and $\mathbf{v}$ (Fig. 10.9). Note that for nonrelativistic velocities ($v^2 \ll c^2$),

$$V(\mathbf{r}, t) \approx \frac{1}{4\pi\epsilon_0} \frac{q}{R}.$$

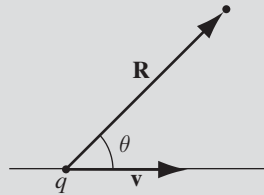


Fig. 10.9

Problem 10.18. I showed that at *most* one point on the particle trajectory communicates with $\mathbf{r}$ at any given time. In some cases there may be *no* such point (an observer at $\mathbf{r}$ would not see the particle – in the colorful language of general relativity, it is “over the **horizon**”). As an example, consider a particle in **hyperbolic motion** along the x -axis:

$$\mathbf{w}(t) = \sqrt{b^2 + (ct)^2} \hat{\mathbf{x}} \quad (-\infty < t < \infty). \quad (10.52)$$

(In special relativity, this is the trajectory of a particle subject to a constant force $F = mc^2/b$.) Sketch the graph of w versus t . At four or five representative points on the curve, draw the trajectory of a light signal emitted by the particle at that point – both in the plus x direction and in the minus x direction. What region on your graph corresponds to points and times (x, t) from which the particle cannot be seen? At what time does someone at point x first see the particle? (Prior to this the potential at x is zero.) Is it possible for a particle, once seen, to *disappear* from view?

- ! **Problem 10.19.** Determine the Liénard–Wiechert potentials for a charge in hyperbolic motion (Eq. 10.52). Assume the point $\mathbf{r}$ is on the x -axis and to the right of the charge ($x > w(t_r)$).¹⁷

¹⁷ The *fields* of a point charge in hyperbolic motion are notoriously tricky. Indeed, a straightforward application of the Liénard–Wiechert potentials yields an electric field in violation of Gauss’s law. This paradox was resolved by Bondi and Gold in 1955. For a history of the problem, see E. Eriksen and Ø. Grøn, *Ann. Phys.* **286**, 320 (2000).

10.3.2 The Fields of a Moving Point Charge

We are now in a position to calculate the electric and magnetic fields of a point charge in arbitrary motion, using the Liénard–Wiechert potentials:¹⁸

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{(rc - \mathbf{r} \cdot \mathbf{v})}, \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{v}}{c^2} V(\mathbf{r}, t), \quad (10.53)$$

and the equations for $\mathbf{E}$ and $\mathbf{B}$:

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

The differentiation is tricky, however, because

$$\mathbf{r} = \mathbf{r} - \mathbf{w}(t_r) \quad \text{and} \quad \mathbf{v} = \dot{\mathbf{w}}(t_r) \quad (10.54)$$

are both evaluated at the retarded time, and t_r – defined implicitly by the equation

$$|\mathbf{r} - \mathbf{w}(t_r)| = c(t - t_r) \quad (10.55)$$

– is *itself* a function of $\mathbf{r}$ and t .¹⁹ So hang on: the next two pages are rough going ... but the answer is worth the effort.

Let's begin with the gradient of V :

$$\nabla V = \frac{qc}{4\pi\epsilon_0} \frac{-1}{(rc - \mathbf{r} \cdot \mathbf{v})^2} \nabla(rc - \mathbf{r} \cdot \mathbf{v}). \quad (10.56)$$

Since $r = c(t - t_r)$,²⁰

$$\nabla r = -c\nabla t_r. \quad (10.57)$$

As for the second term, product rule 4 gives

$$\nabla(\mathbf{r} \cdot \mathbf{v}) = (\mathbf{r} \cdot \nabla)\mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{r} + \mathbf{r} \times (\nabla \times \mathbf{v}) + \mathbf{v} \times (\nabla \times \mathbf{r}). \quad (10.58)$$

Evaluating these terms one at a time,

$$\begin{aligned} (\mathbf{r} \cdot \nabla)\mathbf{v} &= \left(r_x \frac{\partial}{\partial x} + r_y \frac{\partial}{\partial y} + r_z \frac{\partial}{\partial z} \right) \mathbf{v}(t_r) \\ &= r_x \frac{d\mathbf{v}}{dt_r} \frac{\partial t_r}{\partial x} + r_y \frac{d\mathbf{v}}{dt_r} \frac{\partial t_r}{\partial y} + r_z \frac{d\mathbf{v}}{dt_r} \frac{\partial t_r}{\partial z} \\ &= \mathbf{a}(\mathbf{r} \cdot \nabla t_r), \end{aligned} \quad (10.59)$$

where $\mathbf{a} \equiv \dot{\mathbf{v}}$ is the *acceleration* of the particle at the retarded time. Now

$$(\mathbf{v} \cdot \nabla)\mathbf{r} = (\mathbf{v} \cdot \nabla)\mathbf{r} - (\mathbf{v} \cdot \nabla)\mathbf{w}, \quad (10.60)$$

¹⁸ You can get the fields directly from Jefimenko's equations, but it's not easy. See, for example, M. A. Heald and J. B. Marion, *Classical Electromagnetic Radiation*, 3rd ed., Saunders, Orlando, FL (1995), Sect. 8.4.

¹⁹ The following calculation is done by the most direct, “brute-force” method. For a more clever and efficient approach, see J. D. Jackson, *Classical Electrodynamics*, 3rd ed., John Wiley, New York (1999), Sect. 14.1.

²⁰ Remember that $\mathbf{r} = \mathbf{r} - \mathbf{w}(t_r)$ (Fig. 10.8), and t_r is itself a function of $\mathbf{r}$. Contrast Prob. 1.13 (and Section 10.2), where $\mathbf{r} = \mathbf{r} - \mathbf{r}'$ (Fig. 10.3), and $\mathbf{r}'$ was an *independent* variable. In that case $\nabla r = \hat{\mathbf{r}}$, but here we have a more complicated problem on our hands.

and

$$\begin{aligned}
 (\mathbf{v} \cdot \nabla) \mathbf{r} &= \left(v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right) (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) \\
 &= v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}} = \mathbf{v},
 \end{aligned} \tag{10.61}$$

while

$$(\mathbf{v} \cdot \nabla) \mathbf{w} = \mathbf{v}(\mathbf{v} \cdot \nabla t_r)$$

(same reasoning as Eq. 10.59). Moving on to the third term in Eq. 10.58,

$$\begin{aligned}
 \nabla \times \mathbf{v} &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}} \\
 &= \left(\frac{dv_z}{dt_r} \frac{\partial t_r}{\partial y} - \frac{dv_y}{dt_r} \frac{\partial t_r}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{dv_x}{dt_r} \frac{\partial t_r}{\partial z} - \frac{dv_z}{dt_r} \frac{\partial t_r}{\partial x} \right) \hat{\mathbf{y}} \\
 &\quad + \left(\frac{dv_y}{dt_r} \frac{\partial t_r}{\partial x} - \frac{dv_x}{dt_r} \frac{\partial t_r}{\partial y} \right) \hat{\mathbf{z}} \\
 &= -\mathbf{a} \times \nabla t_r.
 \end{aligned} \tag{10.62}$$

Finally,

$$\nabla \times \mathbf{z} = \nabla \times \mathbf{r} - \nabla \times \mathbf{w}, \tag{10.63}$$

but $\nabla \times \mathbf{r} = \mathbf{0}$, while, by the same argument as Eq. 10.62,

$$\nabla \times \mathbf{w} = -\mathbf{v} \times \nabla t_r. \tag{10.64}$$

Putting all this back into Eq. 10.58, and using the “BAC–CAB” rule to reduce the triple cross products,

$$\begin{aligned}
 \nabla(\mathbf{z} \cdot \mathbf{v}) &= \mathbf{a}(\mathbf{z} \cdot \nabla t_r) + \mathbf{v} - \mathbf{v}(\mathbf{v} \cdot \nabla t_r) - \mathbf{z} \times (\mathbf{a} \times \nabla t_r) + \mathbf{v} \times (\mathbf{v} \times \nabla t_r) \\
 &= \mathbf{v} + (\mathbf{z} \cdot \mathbf{a} - v^2) \nabla t_r.
 \end{aligned} \tag{10.65}$$

Collecting Eqs. 10.57 and 10.65, we have

$$\nabla V = \frac{qc}{4\pi\epsilon_0} \frac{1}{(zc - \mathbf{z} \cdot \mathbf{v})^2} [\mathbf{v} + (c^2 - v^2 + \mathbf{z} \cdot \mathbf{a}) \nabla t_r]. \tag{10.66}$$

To complete the calculation, we need to know ∇t_r . This can be found by taking the gradient of the defining equation (Eq. 10.55) – which we have already done in Eq. 10.57 – and expanding ∇z :

$$\begin{aligned}
 -c \nabla t_r = \nabla z &= \nabla \sqrt{\mathbf{z} \cdot \mathbf{z}} = \frac{1}{2\sqrt{\mathbf{z} \cdot \mathbf{z}}} \nabla(\mathbf{z} \cdot \mathbf{z}) \\
 &= \frac{1}{z} [(\mathbf{z} \cdot \nabla) \mathbf{z} + \mathbf{z} \times (\nabla \times \mathbf{z})].
 \end{aligned} \tag{10.67}$$

But

$$(\mathbf{z} \cdot \nabla) \mathbf{z} = \mathbf{z} - \mathbf{v}(\mathbf{z} \cdot \nabla t_r)$$

(same idea as Eq. 10.60), while (from Eqs. 10.63 and 10.64)

$$\nabla \times \mathbf{z} = (\mathbf{v} \times \nabla t_r).$$

Thus

$$-c\nabla t_r = \frac{1}{\mathcal{L}}[\mathbf{r} - \mathbf{v}(\mathbf{r} \cdot \nabla t_r) + \mathbf{r} \times (\mathbf{v} \times \nabla t_r)] = \frac{1}{\mathcal{L}}[\mathbf{r} - (\mathbf{r} \cdot \mathbf{v})\nabla t_r],$$

and hence

$$\nabla t_r = \frac{-\mathbf{r}}{\mathcal{L}c - \mathbf{r} \cdot \mathbf{v}}. \quad (10.68)$$

Incorporating this result into Eq. 10.66, I conclude that

$$\nabla V = \frac{1}{4\pi\epsilon_0} \frac{qc}{(\mathcal{L}c - \mathbf{r} \cdot \mathbf{v})^3} [(\mathcal{L}c - \mathbf{r} \cdot \mathbf{v})\mathbf{v} - (c^2 - v^2 + \mathbf{r} \cdot \mathbf{a})\mathbf{r}]. \quad (10.69)$$

A similar calculation, which I shall leave for you (Prob. 10.20), yields

$$\begin{aligned} \frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0} \frac{qc}{(\mathcal{L}c - \mathbf{r} \cdot \mathbf{v})^3} & \left[(\mathcal{L}c - \mathbf{r} \cdot \mathbf{v})(-\mathbf{v} + \mathbf{r}\mathbf{a}/c) \right. \\ & \left. + \frac{\mathcal{L}}{c}(c^2 - v^2 + \mathbf{r} \cdot \mathbf{a})\mathbf{v} \right]. \end{aligned} \quad (10.70)$$

Combining these results, and introducing the vector

$$\mathbf{u} \equiv c\hat{\mathbf{r}} - \mathbf{v}, \quad (10.71)$$

I find

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{\mathcal{L}}{(\mathbf{r} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \mathbf{r} \times (\mathbf{u} \times \mathbf{a})]. \quad (10.72)$$

Meanwhile,

$$\nabla \times \mathbf{A} = \frac{1}{c^2} \nabla \times (V\mathbf{v}) = \frac{1}{c^2} [V(\nabla \times \mathbf{v}) - \mathbf{v} \times (\nabla V)].$$

We have already calculated $\nabla \times \mathbf{v}$ (Eq. 10.62) and ∇V (Eq. 10.69). Putting these together,

$$\nabla \times \mathbf{A} = -\frac{1}{c} \frac{q}{4\pi\epsilon_0} \frac{1}{(\mathbf{u} \cdot \mathbf{r})^3} \mathbf{r} \times [(c^2 - v^2)\mathbf{v} + (\mathbf{r} \cdot \mathbf{a})\mathbf{v} + (\mathbf{r} \cdot \mathbf{u})\mathbf{a}].$$

The quantity in brackets is strikingly similar to the one in Eq. 10.72, which can be written, using the BAC–CAB rule, as $[(c^2 - v^2)\mathbf{u} + (\mathbf{r} \cdot \mathbf{a})\mathbf{u} - (\mathbf{r} \cdot \mathbf{u})\mathbf{a}]$; the main difference is that we have $\mathbf{v}$'s instead of $\mathbf{u}$'s in the first two terms. In fact, since it's all crossed into $\mathbf{r}$ anyway, we can with impunity *change* these $\mathbf{v}$'s into $-\mathbf{u}$'s; the extra term proportional to $\mathbf{r}$ disappears in the cross product. It follows that

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{r}} \times \mathbf{E}(\mathbf{r}, t). \quad (10.73)$$

Evidently, *the magnetic field of a point charge is always perpendicular to the electric field, and to the vector from the retarded point.*

The first term in $\mathbf{E}$ (the one involving $(c^2 - v^2)\mathbf{u}$) falls off as the inverse *square* of the distance from the particle. If the velocity and acceleration are both zero, this term alone survives and reduces to the old electrostatic result

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{\mathcal{L}^2} \hat{\mathbf{r}}.$$

For this reason, the first term in $\mathbf{E}$ is sometimes called the **generalized Coulomb field**. (Because it does not depend on the acceleration, it is also known as the **velocity field**.) The second term (the one involving $\mathbf{r} \times (\mathbf{u} \times \mathbf{a})$) falls off as the inverse *first* power of r and is therefore dominant at large distances. As we shall see in Chapter 11, it is this term that is responsible for electromagnetic radiation; accordingly, it is called the **radiation field** – or, since it is proportional to a , the **acceleration field**. The same terminology applies to the magnetic field.

Back in Chapter 2, I remarked that if we could write down the formula for the force that one point charge exerts on another, we would be done with electrodynamics, in principle. That, together with the superposition principle, would tell us the force exerted on a test charge Q by any configuration whatsoever. Well, here we are: Eqs. 10.72 and 10.73 give us the fields, and the Lorentz force law determines the resulting force:

$$\mathbf{F} = \frac{qQ}{4\pi\epsilon_0} \frac{r}{(\mathbf{r} \cdot \mathbf{u})^3} \left\{ [(c^2 - v^2)\mathbf{u} + \mathbf{r} \times (\mathbf{u} \times \mathbf{a})] + \frac{\mathbf{V}}{c} \times [\hat{\mathbf{r}} \times [(c^2 - v^2)\mathbf{u} + \mathbf{r} \times (\mathbf{u} \times \mathbf{a})]] \right\}, \quad (10.74)$$

where $\mathbf{V}$ is the velocity of Q , and $\mathbf{r}$, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{a}$ are all evaluated at the retarded time. The entire theory of classical electrodynamics is contained in that equation ... but you see why I preferred to start out with Coulomb's law.

Example 10.4. Calculate the electric and magnetic fields of a point charge moving with constant velocity.

Solution. Putting $\mathbf{a} = \mathbf{0}$ in Eq. 10.72,

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{(c^2 - v^2)\mathbf{r}}{(\mathbf{r} \cdot \mathbf{u})^3}.$$

In this case, using $\mathbf{w} = \mathbf{v}t$,

$$\mathbf{r}\mathbf{u} = c\mathbf{r} - \mathbf{r}\mathbf{v} = c(\mathbf{r} - \mathbf{v}t_r) - c(t - t_r)\mathbf{v} = c(\mathbf{r} - \mathbf{v}t).$$

In Ex. 10.3 we found that

$$\mathbf{r}\mathbf{c} - \mathbf{r} \cdot \mathbf{v} = \mathbf{r} \cdot \mathbf{u} = \sqrt{(c^2t - \mathbf{r} \cdot \mathbf{v})^2 + (c^2 - v^2)(r^2 - c^2t^2)}.$$

In Prob. 10.17 you showed that this radical could be written as

$$Rc\sqrt{1 - v^2 \sin^2 \theta/c^2},$$

where

$$\mathbf{R} \equiv \mathbf{r} - \mathbf{v}t$$

is the vector from the *present* location of the particle to $\mathbf{r}$, and θ is the angle between $\mathbf{R}$ and $\mathbf{v}$ (Fig. 10.9). Thus

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1 - v^2/c^2}{(1 - v^2 \sin^2 \theta/c^2)^{3/2}} \frac{\hat{\mathbf{R}}}{R^2}. \quad (10.75)$$

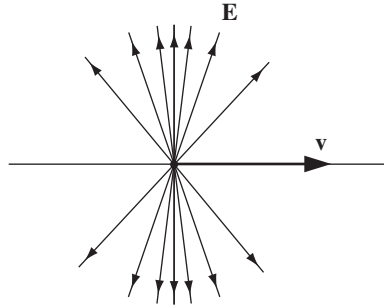


Fig. 10.10

Notice that $\mathbf{E}$ points along the line from the *present* position of the particle. This is an extraordinary coincidence, since the “message” came from the *retarded* position. Because of the $\sin^2 \theta$ in the denominator, the field of a fast-moving charge is flattened out like a pancake in the direction perpendicular to the motion (Fig. 10.10). In the forward and backward directions $\mathbf{E}$ is *reduced* by a factor $(1 - v^2/c^2)$ relative to the field of a charge at rest; in the perpendicular direction it is *enhanced* by a factor $1/\sqrt{1 - v^2/c^2}$.

As for $\mathbf{B}$, we have

$$\hat{\mathbf{z}} = \frac{\mathbf{r} - \mathbf{v}t_r}{r} = \frac{(\mathbf{r} - \mathbf{v}t) + (t - t_r)\mathbf{v}}{r} = \frac{\mathbf{R}}{r} + \frac{\mathbf{v}}{c},$$

and therefore

$$\mathbf{B} = \frac{1}{c}(\hat{\mathbf{z}} \times \mathbf{E}) = \frac{1}{c^2}(\mathbf{v} \times \mathbf{E}). \quad (10.76)$$

Lines of $\mathbf{B}$ circle around the charge, as shown in Fig. 10.11.

The fields of a point charge moving at constant velocity (Eqs. 10.75 and 10.76) were first obtained by Oliver Heaviside in 1888.²¹ When $v^2 \ll c^2$ they reduce to

$$\mathbf{E}(\mathbf{r}, t) \approx \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \hat{\mathbf{R}}, \quad \mathbf{B}(\mathbf{r}, t) \approx \frac{\mu_0}{4\pi} \frac{q}{R^2} (\mathbf{v} \times \hat{\mathbf{R}}). \quad (10.77)$$

The first is essentially Coulomb’s law, and the second is the “Biot–Savart law for a point charge” I warned you about in Chapter 5 (Eq. 5.43).

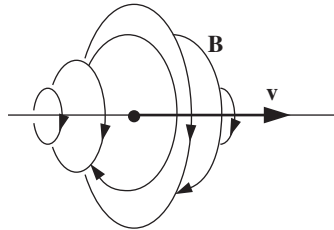


Fig. 10.11

²¹ For history and references, see O. J. Jefimenko, *Am. J. Phys.* **62**, 79 (1994).

Example 10.5. (The Thomson kink model) A point charge is at rest at the origin when, at time $t = 0$, it is given a kick in the x direction, and flies off at constant speed v . Sketch the electric field at some later time. How about the magnetic field?

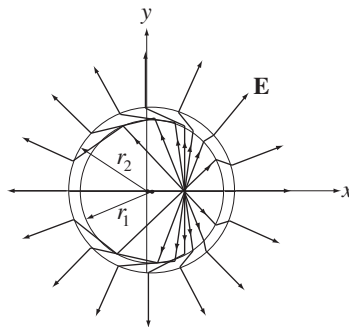


Fig. 10.12

Solution. Let's say the "kick" lasts a short time τ (starting at $t = 0$), during which the particle accelerates up to speed v . Outside a sphere of radius $r_2 = ct$, "news" of the kick has not yet arrived, and the field is that of a particle at rest at the origin (Fig. 10.12). Inside a sphere of radius $r_1 = c(t - \tau)$,²² the news *has* arrived, and $\mathbf{E}$ is the "flattened" field of a particle moving at constant velocity (Fig. 10.10). Within the shell between these spheres we don't know precisely what the field looks like (because we haven't been told exactly how the charge accelerates during the kick); but this much we *do* know: the field lines must join up. Within the shell (which, by the way, is expanding at the speed of light) the electric field has a "kink," and in particular it acquires a significant *tangential* component.

Meanwhile, the *magnetic* field is zero outside the outer sphere, and circles around the x -axis inside the inner sphere (Fig. 10.11). It is everywhere perpendicular to $\mathbf{E}$ (Eq. 10.73). Within the expanding shell the Poynting vector has a substantial radial component: energy is "radiating" away from the charge at the speed of light.²³

Problem 10.20. Derive Eq. 10.70. First show that

$$\frac{\partial t_r}{\partial t} = \frac{rc}{\mathbf{r} \cdot \mathbf{u}}. \quad (10.78)$$

²² The inner sphere is centered at the location of the charge at time $t = \tau$.

²³ This lovely qualitative model for the electromagnetic radiation from an accelerating charge is due originally to J. J. Thomson, *Electricity and Matter*, Yale University Press, New Haven, CT (1904), p. 55. See also E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, 3rd ed., Cambridge University Press (2013), Sect. 5.7 and App. H; R. Y Tsien, *Am. J. Phys.* **40**, 46 (1972); H. C. Ohanian, *Am. J. Phys.* **48**, 170 (1980).

Problem 10.21.

- (a) Suppose a point charge q is constrained to move along the x -axis. Show that the fields at points on the axis to the *right* of the charge are given by

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \left(\frac{c+v}{c-v} \right) \hat{\mathbf{x}}, \quad \mathbf{B} = \mathbf{0}.$$

(Do not assume v is constant!) What are the fields on the axis to the *left* of the charge?

- (b) In Ex. 10.4 I noted that $\mathbf{E}$ is reduced in the forward and backward directions by a factor of $(1 - (v/c)^2)$, in the case of constant velocity. This might appear to contradict your result in (a). Resolve this “paradox.”

Problem 10.22. For a point charge moving at constant velocity, calculate the flux integral $\oint \mathbf{E} \cdot d\mathbf{a}$ (using Eq. 10.75), over the surface of a sphere centered at the present location of the charge.²⁴

Problem 10.23.

- (a) Use Eq. 10.75 to calculate the electric field a distance d from an infinite straight wire carrying a uniform line charge λ , moving at a constant speed v down the wire.
 (b) Use Eq. 10.76 to find the *magnetic* field of this wire.

Problem 10.24. For the configuration in Prob. 10.16, find the electric and magnetic fields at the center. From your formula for $\mathbf{B}$, determine the magnetic field at the center of a circular loop carrying a steady current I , and compare your answer with the result of Ex. 5.6.

More Problems on Chapter 10

Problem 10.25. Suppose you take a plastic ring of radius a and glue charge on it, so that the line charge density is $\lambda_0 |\sin(\theta/2)|$. Then you spin the loop about its axis at an angular velocity ω . Find the (exact) scalar and vector potentials at the center of the ring. [Answer: $\mathbf{A} = (\mu_0 \lambda_0 \omega a / 3\pi) \{ \sin[\omega(t - a/c)] \hat{\mathbf{x}} - \cos[\omega(t - a/c)] \hat{\mathbf{y}} \}.$]

Problem 10.26. Figure 2.35 summarizes the laws of electrostatics in a “triangle diagram” relating the source (ρ), the field ($\mathbf{E}$), and the potential (V). Figure 5.51 does the same for magnetostatics, where the source is $\mathbf{J}$, the field is $\mathbf{B}$, and the potential is $\mathbf{A}$. Construct the analogous diagram for *electrodynamics*, with sources ρ and $\mathbf{J}$ (constrained by the continuity equation), fields $\mathbf{E}$ and $\mathbf{B}$, and potentials V and $\mathbf{A}$ (constrained by the Lorenz gauge condition). Do not include formulas for V and $\mathbf{A}$ in terms of $\mathbf{E}$ and $\mathbf{B}$.

²⁴ Wheeler’s *First Moral Principle* says “Never make a calculation until you know the answer” (E. F. Taylor and J. A. Wheeler, *Spacetime Physics*, Freeman (1966), p. 60). It doesn’t always work, but this is a good problem to try it on.

Problem 10.27. An expanding sphere, radius $R(t) = vt$ ($t > 0$, constant v) carries a charge Q , uniformly distributed over its volume. Evaluate the integral

$$Q_{\text{eff}} \equiv \int \rho(\mathbf{r}, t_r) d\tau$$

with respect to the center. Show that $Q_{\text{eff}} \approx Q(1 - \frac{3v}{4c})$, if $v \ll c$.

Problem 10.28. Check that the potentials of a point charge moving at constant velocity (Eqs. 10.49 and 10.50) satisfy the Lorenz gauge condition (Eq. 10.12).

Problem 10.29. One particle, of charge q_1 , is held at rest at the origin. Another particle, of charge q_2 , approaches along the x -axis, in hyperbolic motion:

$$x(t) = \sqrt{b^2 + (ct)^2};$$

it reaches the closest point, b , at time $t = 0$, and then returns out to infinity.

- What is the force F_2 on q_2 (due to q_1) at time t ?
- What total impulse ($I_2 = \int_{-\infty}^{\infty} F_2 dt$) is delivered to q_2 by q_1 ?
- What is the force F_1 on q_1 (due to q_2) at time t ?
- What total impulse ($I_1 = \int_{-\infty}^{\infty} F_1 dt$) is delivered to q_1 by q_2 ? [Hint: It might help to review Prob. 10.18 before doing this integral. Answer: $I_2 = -I_1 = q_1 q_2 / 4\epsilon_0 b c$.]

Problem 10.30. We are now in a position to treat the example in Section 8.2.1 quantitatively. Suppose q_1 is at $x = -vt$ and q_2 is at $y = -vt$ (Fig. 8.3, with $t < 0$). Find the electric and magnetic forces on q_1 and q_2 . Is Newton's third law obeyed?

Problem 10.31. A uniformly charged rod (length L , charge density λ) slides out the x -axis at constant speed v . At time $t = 0$ the back end passes the origin (so its position as a function of time is $x = vt$, while the front end is at $x = vt + L$). Find the retarded scalar potential at the origin, as a function of time, for $t > 0$. [First determine the retarded time t_1 for the back end, the retarded time t_2 for the front end, and the corresponding retarded positions x_1 and x_2 .] Is your answer consistent with the Liénard–Wiechert potential, in the point charge limit ($L \ll vt$, with $\lambda L = q$)? Do not assume $v \ll c$.

Problem 10.32. A particle of charge q is traveling at constant speed v along the x -axis. Calculate the total power passing through the plane $x = a$, at the moment the particle itself is at the origin. [Answer: $q^2 v / 32\pi\epsilon_0 a^2$.]

Problem 10.33.²⁵ A particle of charge q_1 is at rest at the origin. A second particle, of charge q_2 , moves along the z -axis at constant velocity v .

- Find the force $\mathbf{F}_{12}(t)$ of q_1 on q_2 , at time t (when q_2 is at $z = vt$).
- Find the force $\mathbf{F}_{21}(t)$ of q_2 on q_1 , at time t . Does Newton's third law hold, in this case?

²⁵ See J. J. G. Scanio, *Am. J. Phys.* **43**, 258 (1975).

- (c) Calculate the linear momentum $\mathbf{p}(t)$ in the electromagnetic fields, at time t . (Don't bother with any terms that are constant in time, since you won't need them in part (d)). [Answer: $(\mu_0 q_1 q_2 / 4\pi t) \hat{\mathbf{z}}$.]
- (d) Show that the sum of the forces is equal to minus the rate of change of the momentum in the fields, and interpret this result physically.

Problem 10.34. Develop the potential formulation for electrodynamics with magnetic charge (Eq. 7.53). [Hint: You'll need two scalar potentials and two vector potentials. Use the Lorenz gauge. Find the retarded potentials (generalizing Eqs. 10.26), and give the formulas for $\mathbf{E}$ and $\mathbf{B}$ in terms of the potentials (generalizing Eqs. 10.2 and 10.3).]

- ! **Problem 10.35.** Find the (Lorenz gauge) potentials and fields of a time-dependent ideal electric dipole $\mathbf{p}(t)$ at the origin.²⁶ (It is stationary, but its magnitude and/or direction are changing with time.) Don't bother with the contact term. [Answer:

$$\left. \begin{aligned} V(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2} \cdot [\mathbf{p} + (r/c)\dot{\mathbf{p}}], \\ \mathbf{A}(\mathbf{r}, t) &= \frac{\mu_0}{4\pi} \left[\frac{\dot{\mathbf{p}}}{r} \right], \\ \mathbf{E}(\mathbf{r}, t) &= -\frac{\mu_0}{4\pi} \left\{ \frac{\ddot{\mathbf{p}} - \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})}{r} + c^2 \frac{[\mathbf{p} + (r/c)\dot{\mathbf{p}}] - 3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot [\mathbf{p} + (r/c)\dot{\mathbf{p}}])}{r^3} \right\}, \\ \mathbf{B}(\mathbf{r}, t) &= -\frac{\mu_0}{4\pi} \left\{ \frac{\hat{\mathbf{r}} \times [\dot{\mathbf{p}} + (r/c)\ddot{\mathbf{p}}]}{r^2} \right\}, \end{aligned} \right\} \quad (10.79)$$

where all the derivatives of $\mathbf{p}$ are evaluated at the retarded time.]

- ! **Problem 10.36.** In Section 10.2.1 I noted that the naïve “derivation” of the retarded potentials fails for the fields. The reason is subtle, but it can be fixed.²⁷ In the Lorenz gauge the potentials satisfy the inhomogeneous wave equation (Eq. 10.16):

$$\square^2 f(\mathbf{r}, t) = -s(\mathbf{r}, t), \quad (10.80)$$

where f stands for V or $\mathbf{A}$, and s for the corresponding source (ρ/ϵ_0 or $\mu_0 \mathbf{J}$). The solution (for localized sources) is

$$f(\mathbf{r}, t) = \frac{1}{4\pi} \int \frac{s(\mathbf{r}', t_r)}{r} d\tau' \quad (10.81)$$

(Eq. 10.26), as we guessed, and then confirmed.

- (a) Show (from Maxwell's equations) that

$$\square^2 \mathbf{E} = \frac{1}{\epsilon_0} \nabla \rho + \mu_0 \frac{\partial \mathbf{J}}{\partial t}, \quad \square^2 \mathbf{B} = -\mu_0 (\nabla \times \mathbf{J}). \quad (10.82)$$

²⁶ W. J. M. Kort-Kamp and C. Farina, *Am. J. Phys.* **79**, 111 (2011); D. J. Griffiths, *Am. J. Phys.* **79**, 867 (2011).

²⁷ N. A. Lemos, *Rev. Bras. Ensino Fis.*, **27**, 385 (2005); available in English at arXiv:physics/0508047v1 [physics.ed-ph].

- (b) Write down the solutions for $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B}(\mathbf{r}, t)$.
 (c) Using integration by parts, recover Jefimenko's equations (10.36 and 10.38).

Problem 10.37. Two charges $\pm q$ approach the origin at constant velocity from opposite directions along the x -axis. They collide and stick together, forming a neutral particle at rest. Sketch the electric field before and shortly after the collision. How would you interpret the field after the collision, physically?

- ! **Problem 10.38.** Imagine an ideal electric dipole at the origin, pointing in the z direction, whose dipole moment is increasing linearly with time:

$$\mathbf{p}(t) = kt\hat{\mathbf{z}} \quad (10.83)$$

(for some constant k). Evidently, there is a steady current flowing from the negative end to the positive end:

$$\mathbf{J}(\mathbf{r}) = \alpha\delta^3(\mathbf{r})\hat{\mathbf{z}}, \quad (10.84)$$

where α is a constant to be determined.

- (a) From the continuity equation, find the charge density $\rho(\mathbf{r}, t)$. Stipulate that there is no static background charge: $\rho(\mathbf{r}, 0) = 0$. [Express your answer as a derivative of the delta function.]
 (b) From the charge density, calculate the electric dipole moment, and, comparing Eq. 10.83, determine α (in terms of k). [Hint: Use integration by parts.]
 (c) When $\rho(\mathbf{r}, t)$ is linear in t , and $\mathbf{J}(\mathbf{r})$ is constant, a lucky cancelation means that the electric field satisfies Coulomb's law, and the magnetic field satisfies the Biot–Savart law, with *no retardation* (see Probs. 10.13 and 10.14). In our case, then, $\mathbf{E}(\mathbf{r}, t)$ is given by the electrostatic formula (Eq. 3.106), with $\mathbf{p}$ the *present* dipole moment (Eq. 10.83). What is the magnetic field, $\mathbf{B}(\mathbf{r}, t)$?
 (d) Now suppose there is a *second* such dipole, located on the z -axis at $z = a$ and pointing in the x direction. What is the net force of $\mathbf{p}_1$ on $\mathbf{p}_2$, and of $\mathbf{p}_2$ on $\mathbf{p}_1$? Is Newton's third law obeyed?
 (e) Now the hard part: calculate the field momentum (Eq. 8.29) associated with their interaction,

$$\mathbf{p}_{\text{em}}(t) = \epsilon_0 \int [(\mathbf{E}_1 \times \mathbf{B}_2) + (\mathbf{E}_2 \times \mathbf{B}_1)] d\tau, \quad (10.85)$$

and show that its rate of change exactly balances the net force on the two dipoles, so the *total* momentum is conserved.²⁸ [Use a computer to help with the integrals.]

²⁸ This is a simplified version of O. Narayan, *Am. J. Phys.* **89**, 1033 (2021), due to V. Onoichin. (The self-forces and self-field momenta are all zero.)